Dense critical graphs from saturated graphs

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

Let $f_k(n)$ denote the maximum number of edges in a k-critical graph on n vertices. For every fixed integer $k \geq 8$, we construct a sequence of k-critical graphs G_s with $|V(G_s)| \to \infty$ and

$$\frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \begin{cases} \dfrac{k-4}{2(k-2)}, & k \text{ even,} \\ \dfrac{k-5}{2(k-3)}, & k \text{ odd.} \end{cases}$$

For every $k \geq 8$ except $k = 9$, this improves the best known lower bounds, due to Toft (1970). For k divisible by three and at least twelve, it disproves the asymptotic formula proposed in Erdős Problem 917, previously known to fail when $3 \nmid k$. For $k = 12$, the density is $2/5 > 3/8$. The construction combines a coloring module of Pegden with saturated graphs of sublinear maximum degree. The complete proof is formalized and verified in Lean 4; generative AI was used in developing the arguments and the manuscript.

1 Introduction

All graphs are finite and simple. A graph is k-critical if it has chromatic number k and every proper subgraph is $(k-1)$-colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the maximum number of edges of a k-critical graph on n vertices (in the edge-deletion sense), or 0 if none exists. Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2}\left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2. \tag{1.1}$$

This is the general asymptotic question in Erdős Problem 917 [4]. Dirac [2, p. 90] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős [3, p. 28, eq. (3)] observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [10] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [8, Section 5.1, pp. 97–98] and Jensen [7]. Luo, Ma, and Yang [9] noted that no constructions improving Toft's asymptotic constants had been found since 1970. We give the following construction.

Theorem 1. *For every integer $k \geq 8$, there is a sequence of k-critical graphs G_s such that $|V(G_s)| \to \infty$ and*

$$\frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \begin{cases} \dfrac{k-4}{2(k-2)}, & k \textit{ even,} \\ \dfrac{k-5}{2(k-3)}, & k \textit{ odd.} \end{cases}$$

Write c_k for the density in Theorem 1. Toft's coefficients [10], in the notation of [9, Section 1], are

$$b_k = \frac{1}{2} - \frac{3}{2k - \delta_k}, \qquad \delta_k = \begin{cases} 0, & k \equiv 0 \pmod 3, \\ 8/7, & k \equiv 1 \pmod 3, \\ 44/23, & k \equiv 2 \pmod 3. \end{cases}$$

Corollary 2. *For every $k \geq 8$, $k \neq 9$, the density c_k exceeds Toft's coefficient b_k, and* (1.1) *fails. In particular,* (1.1) *fails for every multiple of three at least twelve. For $k = 12$,*

$$\limsup_{n\to\infty} \frac{f_{12}(n)}{n^2} \geq \frac{2}{5} > \frac{3}{8}.$$

For $k = 9$, the density $c_9 = 1/3$ equals both coefficients.

Proof. Theorem 1 and $f_k(|V(G_s)|) \geq e(G_s)$ give $\limsup_{n\to\infty} f_k(n)/n^2 \geq c_k$. For even $k \geq 8$,

$$2\lfloor k/3 \rfloor < k - 2, \qquad k > 6 - \delta_k,$$

which imply $c_k > \frac{1}{2}(1 - 1/\lfloor k/3 \rfloor)$ and $c_k > b_k$, respectively. For odd $k \geq 11$, the corresponding inequalities are $2\lfloor k/3 \rfloor < k - 3$ and $k > 9 - \delta_k$. Formula (1.1) would force the limit, and hence the limsup, to equal $\frac{1}{2}(1 - 1/\lfloor k/3 \rfloor)$, contradicting the lower bound. Taking $k = 12$ gives the displayed inequality. At $k = 9$, all three coefficients equal $1/3$. $\square$

The failure of (1.1) for $3 \nmid k$ was already implied by Toft's constructions; the new cases are $3 \mid k$, $k \geq 12$.

For even $k = 2t + 2$, the graph G consists of t copies of Pegden's module $U(S, T)$ [11], with their active sets joined by an almost complete t-partite graph. The omitted edges form a graph Z whose adjacency is prescribed by a K_t-saturated graph H of small maximum degree from [1]. In a $(2t+1)$-coloring, each active set needs three colors (Lemma 3); the degree bound forces two of them to be private to that part. The remaining color must then appear in all t parts, giving a K_t in Z, impossible since H is K_t-free. Deleting an edge between two parts adds an edge to Z, and saturation supplies a K_t through it, allowing one color to be shared in a $(2t+1)$-coloring of $G - e$. Complete joins of critical graphs, as in Dirac's construction, cannot share colors between factors; omitting Z costs $o(n^2)$ edges and permits this shared color.

At $t = 2$, the graph H is edgeless and the two active sets are completely joined, as in Pegden's two-module construction [11, Section 2.1]. The limiting density is $1/4$, giving no improvement for $k = 6$ or for $k = 7$ after taking a cone.

Deleting the $O_k(n)$ module edges leaves a t-partite graph, so the asymptotic density of this construction is at most $(t-1)/(2t)$, equal to $2/5$ when $k = 12$. For fixed $k \geq 4$ and sufficiently large n, Luo, Ma, and Yang [9, Theorem 1.1] give

$$f_k(n) \leq e(T_{k-2}(n)) - \frac{n^2}{36(k-1)^2} \leq \left(\frac{1}{2} - \frac{1}{2(k-2)} - \frac{1}{36(k-1)^2}\right) n^2,$$

where $T_{k-2}(n)$ is the balanced complete $(k-2)$-partite graph. For even k, the difference between this upper coefficient and c_k is $1/(2(k-2)) - 1/(36(k-1)^2)$. At $k = 12$, the upper coefficient is $9/20 - 1/4356$. Section 4 constructs the saturated graphs over finite fields and proves Theorem 1.

2 A coloring module

Let $r \geq 3$, let S be r-critical, and let T be $(r-1)$-critical. The module $U(S,T)$ consists of disjoint copies of S and T, together with an independent set

$$A = V(S) \times V(T).$$

Each $(x,y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T, and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x.

The following is the $U(r,r-1)$ case of Pegden's module lemma [11, Lemma 2.5], stated there for triangle-free factors; the lemma holds for general critical factors S and T of the indicated chromatic numbers.

Lemma 3. *Fix a palette of r colors.*

1. *In every proper coloring of $U(S,T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ, there is a proper coloring in which the active set uses only α, β, γ, and γ occurs on the active set only at a_0.*
3. *Given an edge e of $U(S,T)$ and distinct colors α, β, there is a proper coloring of $U(S,T) - e$ in which the active set uses only α, β.*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all r colors, it contains a vertex colored α and a vertex colored β. Every active vertex in the first row must have color β, so no vertex of T can have color β. The second row similarly excludes α from T. This leaves only $r-2$ colors for T, a contradiction.

For (2), color S so that α occurs only at x_0. Color T with the $r-1$ colors other than α, so that β occurs only at y_0. These colorings exist by criticality: delete the specified vertex, color the remaining graph with one fewer color, and give the deleted vertex its own color. Extend these colorings by

$$\mathrm{col}(x,y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0,\ y \neq y_0, \\ \gamma, & (x,y) = (x_0, y_0). \end{cases} \tag{2.1}$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the $r-1$ colors other than α, and color all of A with α. If $e \in E(T)$, color $T - e$ with the $r-2$ colors other than α, β. Choose $x_0 \in V(S)$ and color S with α only at x_0. Color the row indexed by x_0 with β and all other rows with α.

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0, use (2.1) with the color of (x_0, y_0) changed to α. If it joins (x_0, y_0) to y_0, change that color to β. In either case the only edge that would become monochromatic is the deleted edge. □

3 From saturated graphs to critical graphs

A graph is K_t-saturated if it contains no K_t, but adding any missing edge creates a K_t. Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a K_{t-2}.

Proposition 4. *Let $t \geq 3$, and let H be a K_t-saturated graph on $v \geq t$ vertices with maximum degree $d < v - 1$. Let $h \geq 2t + 1$ be odd and suppose*

$$v > 2t(t-1)hd. \tag{3.1}$$

Put $a = 2hv$. There is a $(2t+2)$-critical graph G with

$$|V(G)| = t(a + h + 2v) \tag{3.2}$$

and

$$e(G) \geq \binom{t}{2} a^2 \left(1 - \frac{d}{v}\right). \tag{3.3}$$

Proof. Construction. Set $\ell = 2v$ and take joins of cliques and odd cycles, as in [2, p. 90]:

$$S = K_{2t-2} \vee C_{h-2t+2}, \qquad T = K_{2t-3} \vee C_{\ell-2t+3}.$$

Both cycles are odd and have length at least three. Since the join of a p-critical and a q-critical graph is $(p+q)$-critical, S is $(2t+1)$-critical and T is $2t$-critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take t disjoint copies $U_1, \ldots, U_t$ of $U(S, T)$, with active sets $A_1, \ldots, A_t$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0, 1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z. Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a t-partite graph Z on $A_1 \cup \cdots \cup A_t$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \quad \Longleftrightarrow \quad \pi(u)\pi(w) \in E(H).$$

The graph G retains all the edges of the modules and, between different active sets, takes the complement of Z. There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G.

We record three properties of Z. First, Z is K_t-free: the labeling map sends every clique injectively to a clique of H. Moreover, adding any missing edge between different parts of Z creates a K_t with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a K_{t-2} in $N_H(z) \cap N_H(z')$. Place its labels in the other $t - 2$ parts. Together with u, w, they form the required clique in $Z + uw$. If the labels coincide, say both equal z, choose a vertex $z' \neq z$ not adjacent to z; this is possible because $d < v - 1$. The same common-neighborhood clique is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_{t-1} across any prescribed $t-1$ parts. Indeed, H has a missing edge; one endpoint together with a K_{t-2} in the common neighborhood gives a K_{t-1} in H. Its labels can be placed in any $t - 1$ parts of Z.

Third, $D := \Delta(Z) = (t-1)bd = 2h(t-1)d$, so (3.1) gives $\ell > 2tD$.

The chromatic lower bound. Suppose that G has a $(2t+1)$-coloring. Fix a part A_i and any color α. Since its copy of S uses all $2t + 1$ colors, some vertex x of S has color α. The ℓ active vertices in the row indexed by x avoid α, so the pigeonhole principle and $\ell > 2tD$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i, each used more than D times in A_i.

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the $2t$ chosen colors $\alpha_1, \beta_1, \dots, \alpha_t, \beta_t$ are distinct, leaving a single color γ. The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 3(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_t in Z, a contradiction. Thus $\chi(G) \geq 2t + 2$.

Colorings after edge deletion. Use the $(2t+1)$-color palette

$$\alpha_1, \beta_1, \dots, \alpha_t, \beta_t, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G. By the first property of Z, there is a K_t in $Z + uw$ containing uw, with vertices $a_i \in A_i$. Apply Lemma 3(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i. The only color shared by different active sets is γ. Its t vertices are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained a $(2t+1)$-coloring of $G - e$.

Now let e lie within U_i. Apply Lemma 3(3) to color $U_i - e$, using only α_i, β_i on A_i. Choose a K_{t-1} in Z across the other $t - 1$ parts. In each of those modules, apply Lemma 3(2) with the selected vertex as the only active vertex of color γ. Once again this is a proper $(2t+1)$-coloring of $G - e$.

These cases cover every edge of G. Recoloring one endpoint of a deleted edge with a new color gives $\chi(G) \leq 2t + 2$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is $(2t+2)$-critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $a^2(1 - d/v)$ edges remain between each pair of active sets, proving (3.3). $\square$

4 Saturated graphs of small maximum degree

Hanson and Seyffarth [6] constructed maximal triangle-free graphs of maximum degree $O(\sqrt{n})$, and Füredi and Seress [5] improved the constant. We use [1, Section 4, Example 2], with $k = t$ and one added vertex per line, under the sufficient condition $Q \geq 2t$. The case $t = 3$ is [5, Example 2.2]; its core has no edges between different levels. We give the saturation verification and exact degree counts below.

Lemma 5 (after [1, Example 2])**.** *Let $t \geq 3$ and let $Q \geq 2t$ be a prime power. There is a K_t-saturated graph $H_{t,Q}$ with*

$$\begin{aligned} |V(H_{t,Q})| &= (t-1)(t-2)(Q^2+Q) + Q^2, \\ \Delta(H_{t,Q}) &= \begin{cases} 2Q + 2, & t = 3, \\ Q((t-2)(2t-3)+1) - (t-2), & t \geq 4. \end{cases} \end{aligned} \tag{4.1}$$

In particular, for fixed t, the order is $\Theta_t(Q^2)$ and the maximum degree is $O_t(Q)$.

Proof. Let F be a field of order Q. The levels are $F \cup \{\infty\}$, given any linear order, and the places in each level are indexed by $\mathbb{Z}/Q\mathbb{Z}$ through a fixed bijection with F. For each $(a, b) \in F^2$, form a line $L_{a,b}$ with coordinate $au + b$ in finite level u and coordinate a in level ∞. Every point lies on Q

lines. Two distinct lines meet, and two points in different levels lie on a line; these assertions follow by solving the corresponding linear equations over F.

First suppose $t \geq 4$, and put $m = t - 2$. A core vertex is a quadruple (i, j, τ, c), where i is its level, $j \in \mathbb{Z}/Q\mathbb{Z}$ its place, $\tau \in \mathbb{Z}/m\mathbb{Z}$ its type, and $c \in \{0, \ldots, m\}$ its copy. In one level, adjacency means that the copies differ and the (j, τ) pairs differ. Between levels $i < i'$, adjacency means

$$\tau' = \tau + 1, \qquad j' \in \{j + 1, \ldots, j + m\}.$$

Add an independent vertex for each line, adjacent to all $m(m+1)$ core vertices at each of its points. Place arithmetic is modulo Q, irrespective of the characteristic of F.

Excluding K_t. A core triangle cannot occupy three levels: its edges would require both $\tau'' = \tau+1$ and $\tau'' = \tau + 2$ modulo $m \geq 2$. A clique in one level has at most $m + 1$ vertices. For a clique meeting two levels, the types in each level agree, and its place sets A, B satisfy $B - A \subseteq \{1, \ldots, m\}$ (mod Q). We claim that $|A| + |B| \leq m + 1$. Fix $b_0 \in B$, write $A = \{b_0 - s : s \in S\}$ with $S \subseteq \{1, \ldots, m\}$, and let $s_0 = \min S$. Relative to $a_0 = b_0 - s_0$, write $A = -R$ and $B = J$, where $R = S - s_0 \subseteq \{0, \ldots, m - 1\}$ and $J \subseteq \{1, \ldots, m\}$. For every $d \in R$ and $j \in J$, the integer $d + j$ lies between 1 and $2m - 1 < Q$; its residue belongs to $\{1, \ldots, m\}$, so $d + j \leq m$. Thus

$$|A| + |B| = |R| + |J| \leq (\max R + 1) + \max J \leq m + 1.$$

A clique in a line neighborhood has at most m vertices in one level, since the places agree and the types must differ. If it meets two levels, the same-type observation permits at most one vertex in each. Thus every line neighborhood is K_{m+1}-free, and the whole graph is K_t-free.

Saturation. We exhibit a common K_m for each missing edge xy. For two line vertices, take their intersection point and m core vertices there of distinct types and copies. For a core vertex and a nonincident line, take all m types at the line's point in the core vertex's level, using the m other copies.

For two core vertices in one level with equal copies, take a third place and all m types there, using the other copies. If their copies differ, nonadjacency forces their places and types to agree. Take a line through that point and $m - 1$ core vertices there, using the other types and copies.

Finally, let x lie below $y = (i', j', \tau', c')$. In the level of x, take the m places $j' - 1, \ldots, j' - m$, of type $\tau' - 1$, in the copies other than that of x. They form a K_m adjacent to y. None has the same place and type as x, since that would make x adjacent to y; hence all are also adjacent to x.

The case $t = 3$. Use two copies at each point, with no type coordinate and no edges between core vertices in different levels. Within one level, join vertices exactly when both their places and their copies differ. Add the line vertices as before. The core is bipartite and every line neighborhood is independent, so the graph is triangle-free. Two line vertices have a common core neighbor at their intersection. A core vertex and a nonincident line have a common neighbor at the line's point in the same level, in the other copy. Two core vertices in one level and one copy have a common neighbor at a third place in the other copy; if their copies differ, a missing edge means that their places agree, and a line through that point is a common neighbor. Core vertices in different levels have a common line neighbor. Thus the graph is K_3-saturated.

Order and degree. For $t \geq 4$, there are $m(m + 1)Q(Q + 1)$ core vertices and Q^2 line vertices. A core vertex has $m(mQ - 1)$ neighbors in its level, $m(m + 1)Q$ in other levels, and Q line neighbors. A line vertex has degree $m(m + 1)(Q + 1)$, no larger than the core degree for $Q \geq 2t$. For $t = 3$, the core and line degrees are $2Q - 1$ and $2Q + 2$, respectively. This proves (4.1). $\square$

Proof of Theorem 1. First let $k = 2t + 2$ be even, where $t \geq 3$ is fixed. For integers s sufficiently large that $Q = 9^s \geq 2t$, set

$$h = 3^s, \qquad Q = h^2, \qquad H = H_{t,Q}$$

using Lemma 5. Its order and maximum degree satisfy $v = \Theta_t(h^4)$ and $d = O_t(h^2)$, so $h \to \infty$ and $2t(t-1)hd/v \to 0$. Consequently $h \geq 2t+1$, $v \geq t$, $d < v-1$, and (3.1) hold for all sufficiently large s.

Apply Proposition 4, and write $a = 2hv$ and $N = |V(G)|$. We have

$$\frac{N}{ta} = 1 + \frac{1}{h} + \frac{1}{2v} \longrightarrow 1, \qquad \frac{d}{v} \longrightarrow 0.$$

Since $e(S) = O_t(h)$, $e(T) = O_t(v)$, and each module has $2a$ edges incident with its active set, the edges within modules number $O_t(a)$. Hence

$$\binom{t}{2} a^2 \left(1 - \frac{d}{v}\right) \leq e(G) \leq \binom{t}{2} a^2 + O_t(a).$$

It follows that

$$\frac{e(G)}{N^2} \longrightarrow \frac{\binom{t}{2}}{t^2} = \frac{t-1}{2t} = \frac{k-4}{2(k-2)}.$$

For odd $k = 2t+3 \geq 9$, join each graph in the even construction to K_1. The resulting graph is k-critical, has $N+1$ vertices and $e(G)+N$ edges, and has the same limiting density, now equal to $(k-5)/(2(k-3))$. $\square$

Code availability

Theorem 1 is formalized for every $k \geq 8$ in the top-level declaration `Erdos917.dense_critical_graphs`, available at `https://github.com/FireflySentinel/erdos-917`. Its statement uses the graph-theoretic definitions `IsCritical` and `edgeCount`. Its axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`. Corollary 2 is formalized as `Erdos917.corollary_two`, using `fk`, `erdosCoefficient`, and `toftCoefficient` for the extremal function and the two comparison coefficients; its $k = 12$ instance is `Erdos917.f12_limsup_ge_two_fifths`.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the construction of critical graphs and the density estimates, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

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